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Faculty of Computer Science and Information Technology

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Subject Code: TMF1913/TMF1014	**Subject Name: System Analysis and Design**

Assignment/Project:	**Assignment: (Case Study Number)/Group Lecture**

Name of Lecturer:	**Dr Mohammad bin Hossin**

Due Date: JAN 2025	**Date Submitted:**

All work must be submitted by the due date. If an extension of work is granted, an assignment extension acknowledgement slip must be signed by the lecturer/tutor and attached to assignment.

Please note that is your responsibility to retain copies of your assignment.

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MARK :

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1.Project Summary

This project focuses on addressing the challenges faced by the current food delivery system, including delays, lack of real-time tracking, limited feedback mechanisms and difficulties in communication between users and restaurants.

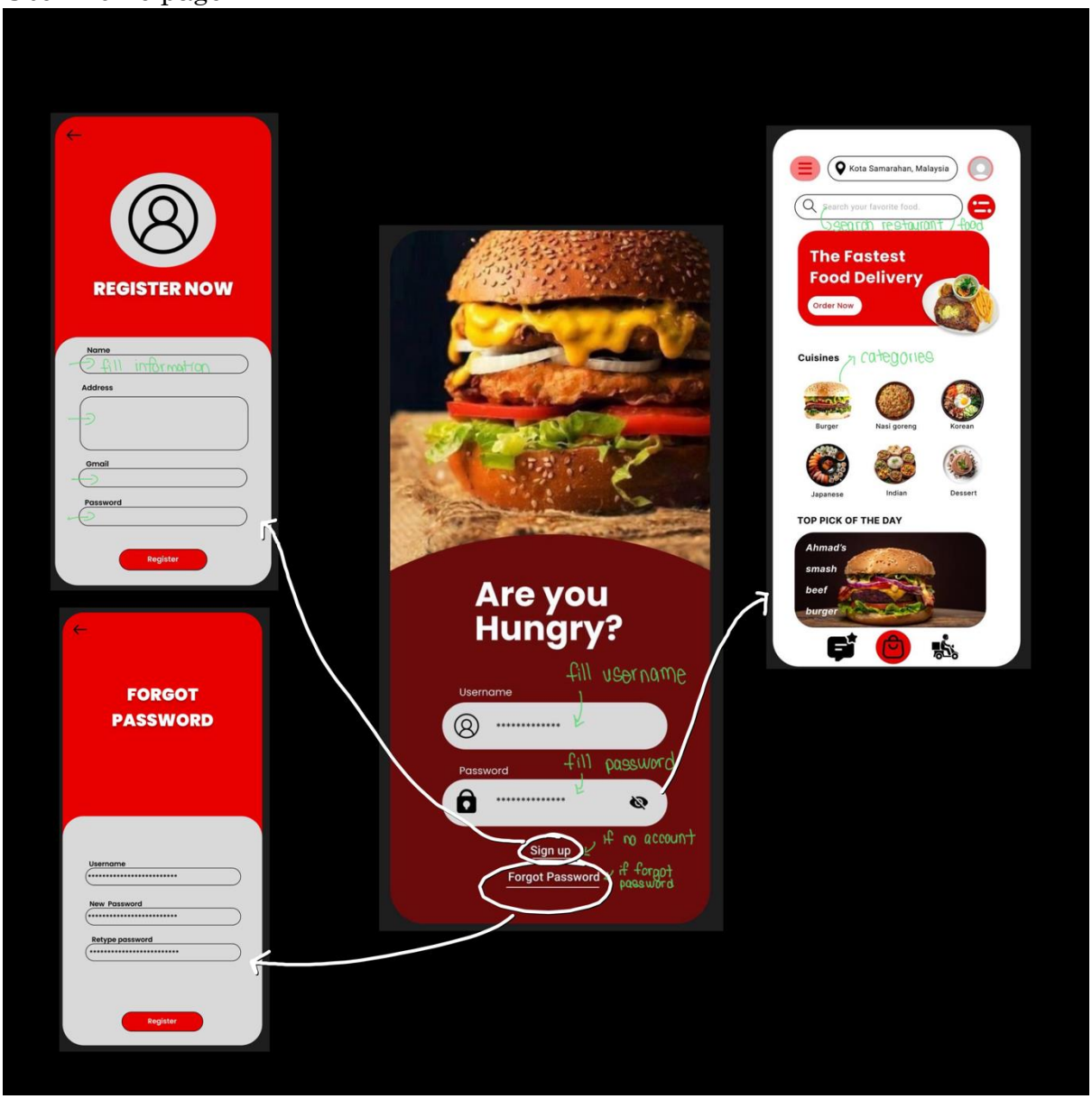
To overcome the challenges, we proposed a modern comprehensive online food delivery application. The app will integrate various functionalities to cater to the needs of customers, restaurants and delivery personnel by offering features such as real-time tracking, efficient order management, secure payment options and robust feedback system. By prioritizing user needs, the system will provide a smooth and convenient ordering process to customers, operational efficiency for restaurants and streamlined logistics for delivery personnel.

The key objectives include ensuring user-friendly functionality, enabling scalability to accommodate future growth and maintaining data and payment security. This app will allow users to browse menus, place orders and monitor delivery process in real-time while offering tools for the restaurants to manage orders effectively. Drivers will benefit from optimized route management and clear communication channels.

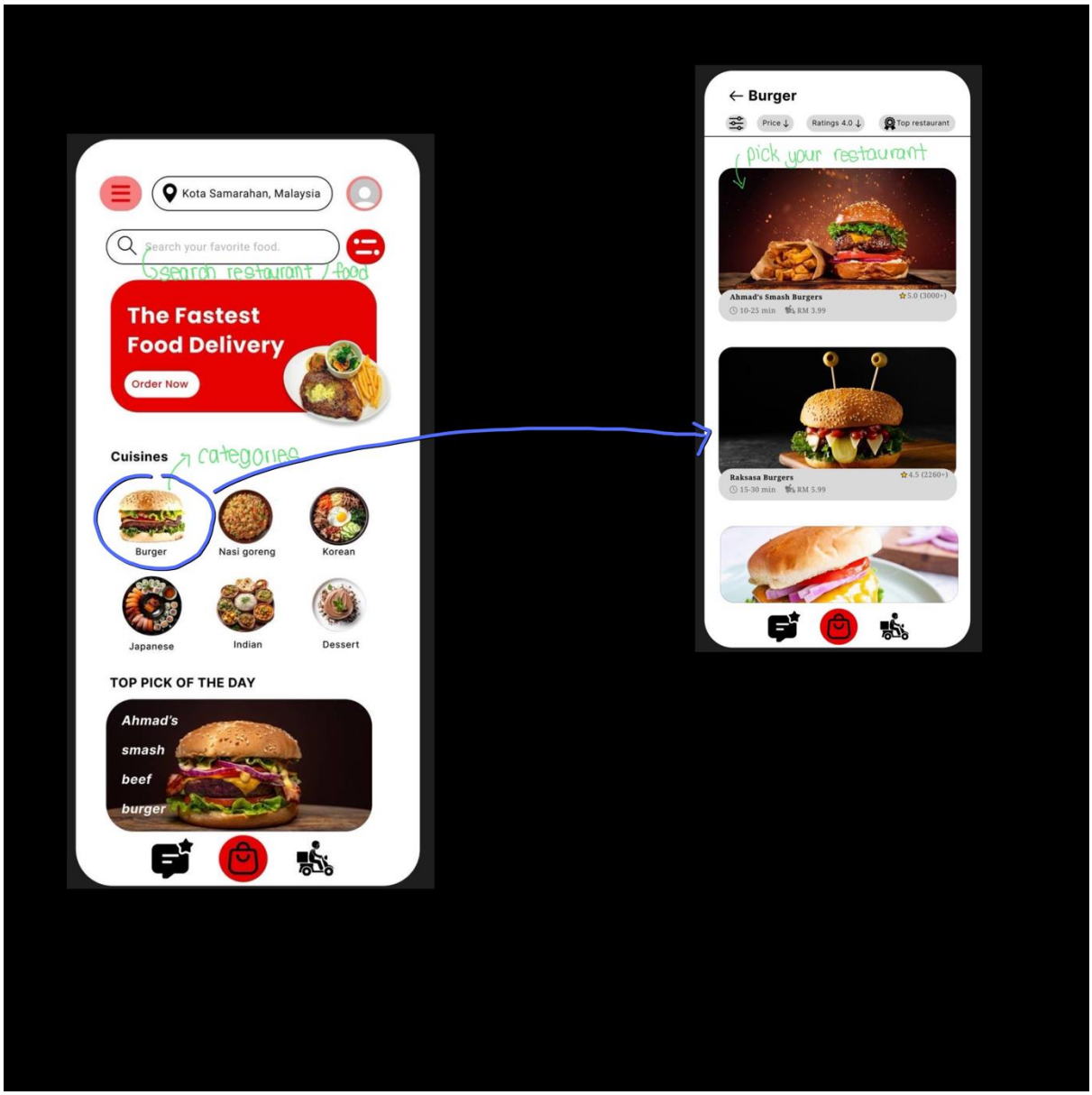
2. User Interface Design

Prototype's link:
https://www.figma.com/proto/b0H1q2ycOS81Yoos7mPiR9/Food-delivery-app?node-id=1-34&p=f&t=UdmETmzJkGFhqPGH-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A34

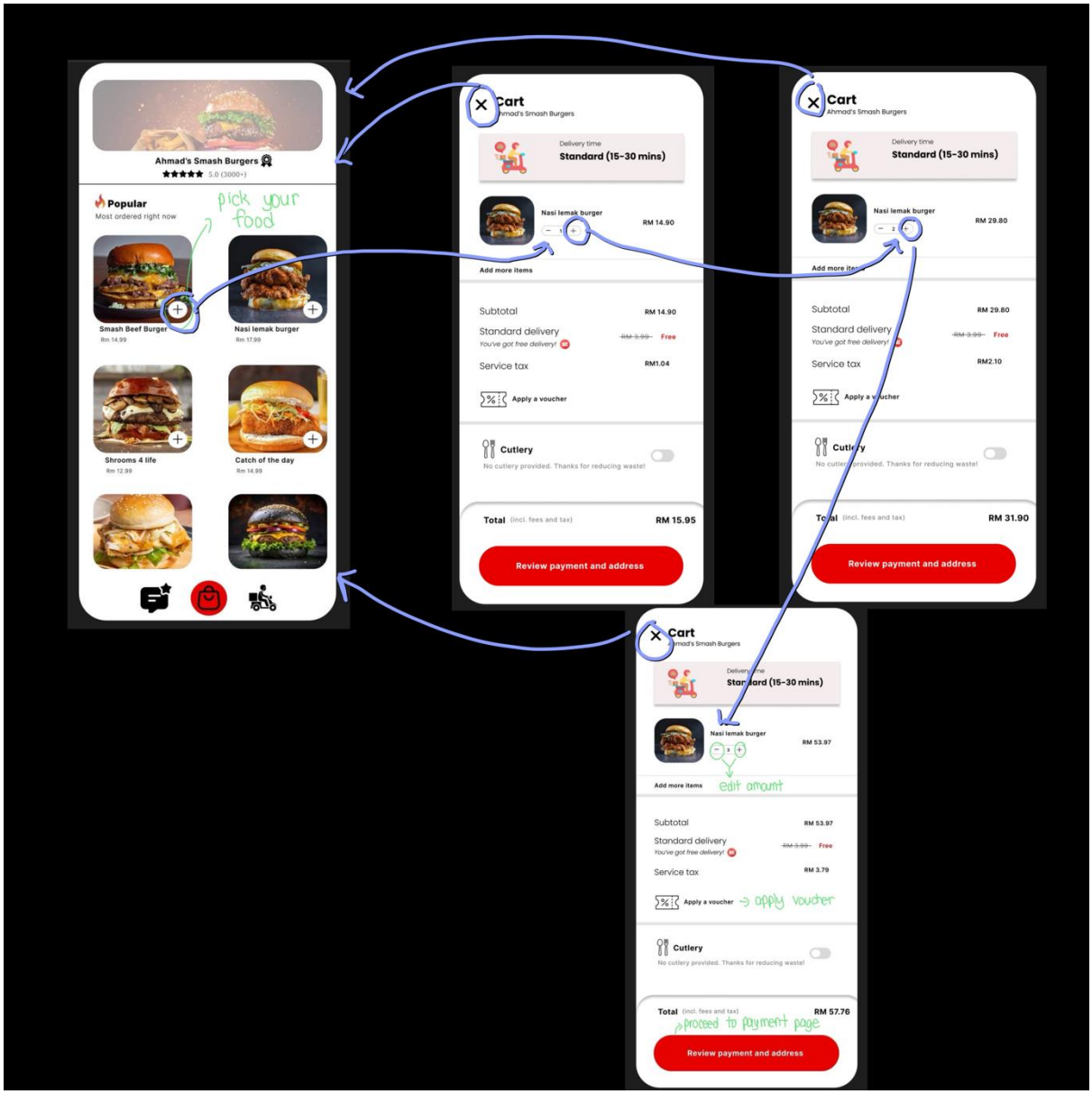
- User/Admin/Restaurant/Delivery Personnel sign in page
- User Home page



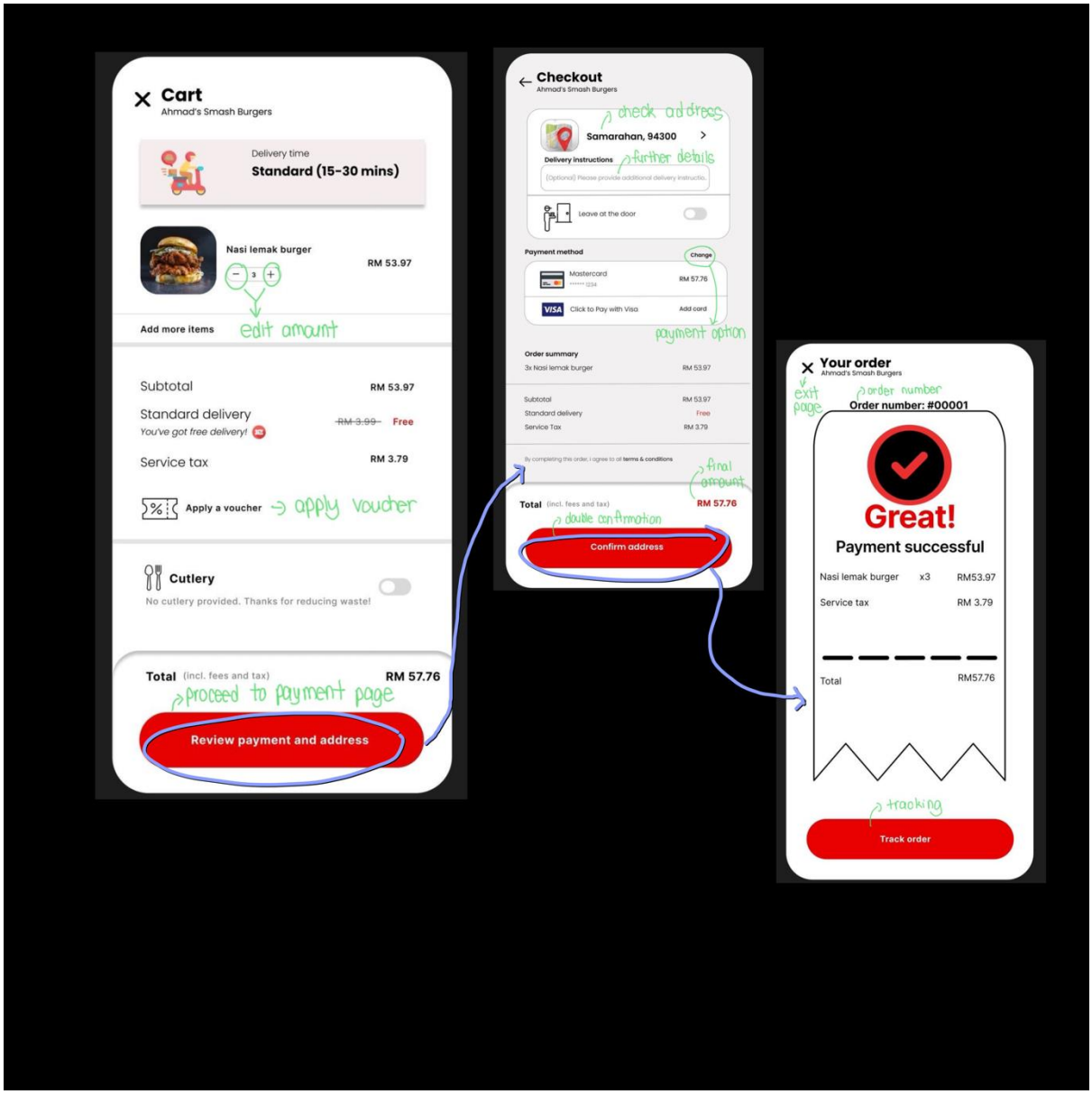
- User Home Page
- Restaurant Selection page



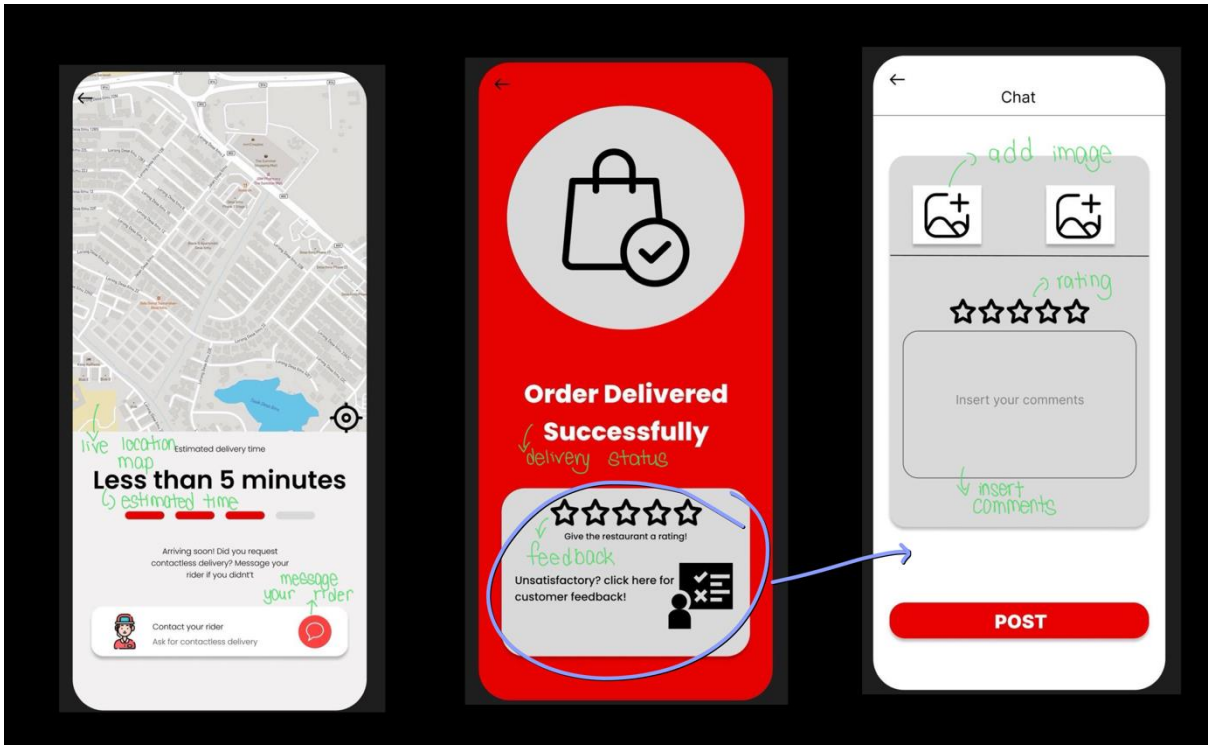
- Food selection page
- First confirmation page



- Second confirmation page
- Receipt



- Tracking page
- Order Success page
- Feedback page



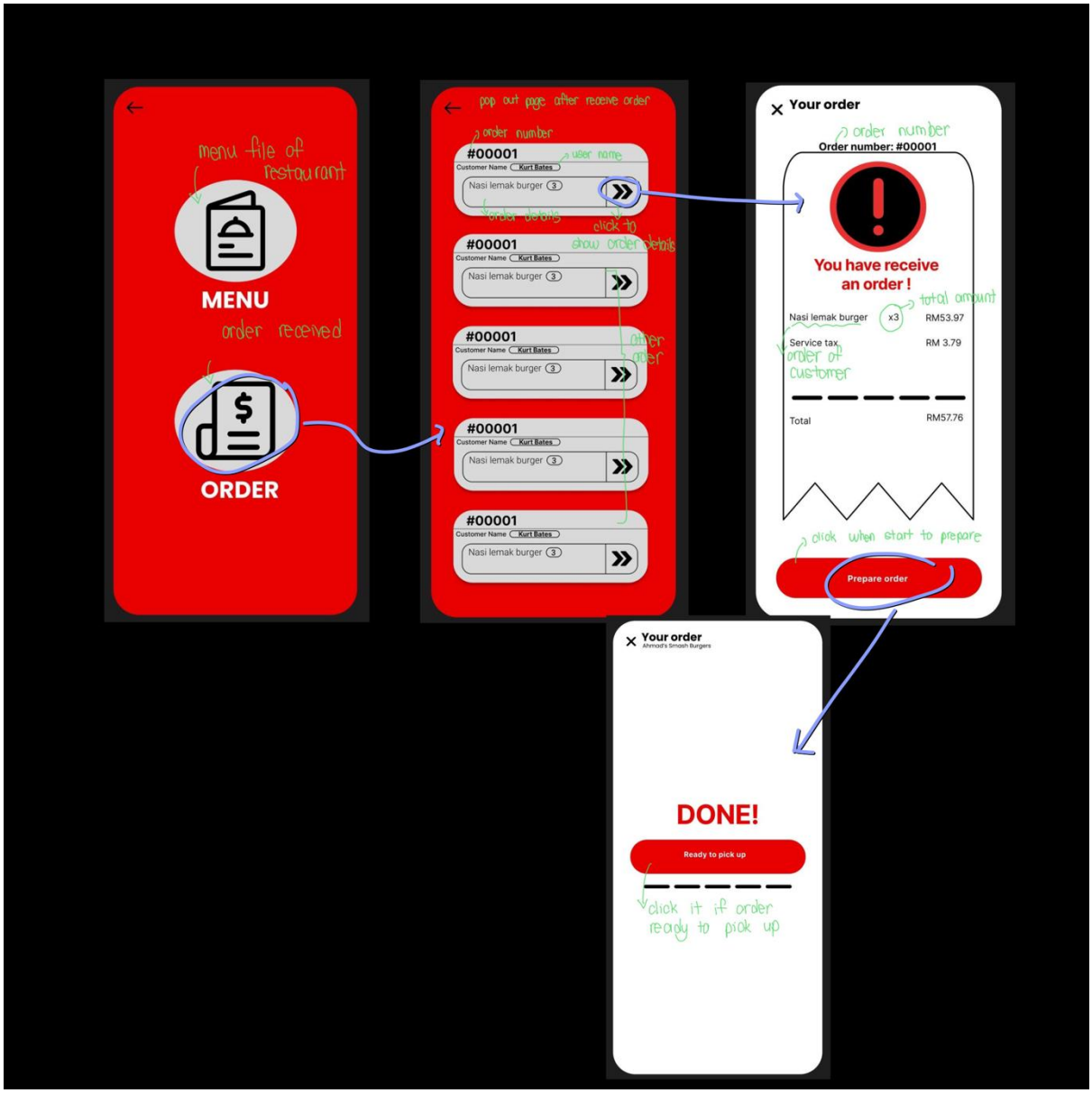
- Admin Home page
- Restaurant/Delivery personnel registration approval
- Feedback file



- Restaurant registration page
- Restaurant Home page
- Update menu page



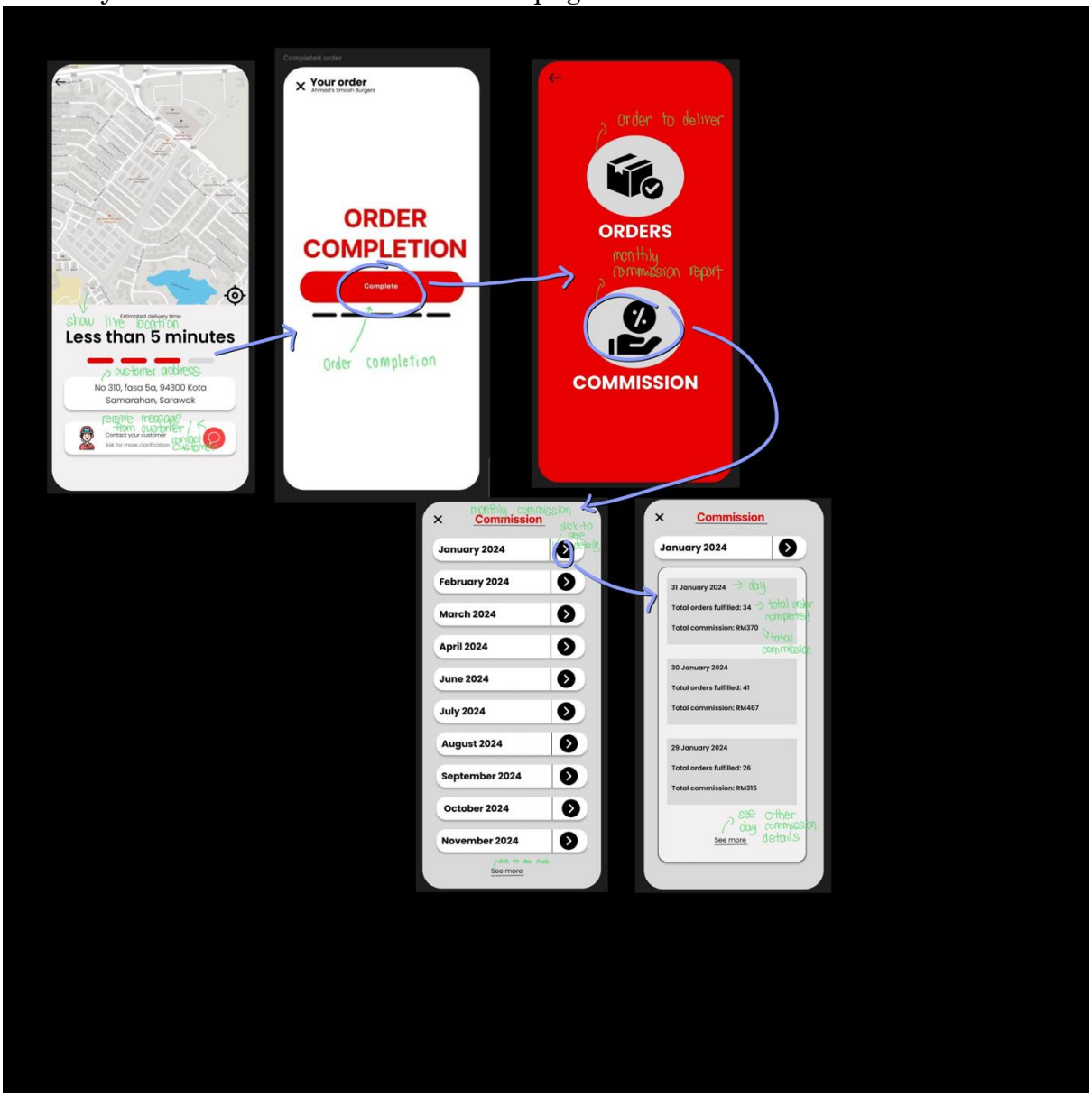
- Receive order page
- Order details page
- Product ready page



- Delivery Personnel registration page
- Delivery Personnel Home page
- Delivery Personnel order page
- Delivery Personnel Order details page
- Delivery Personnel product pick up page

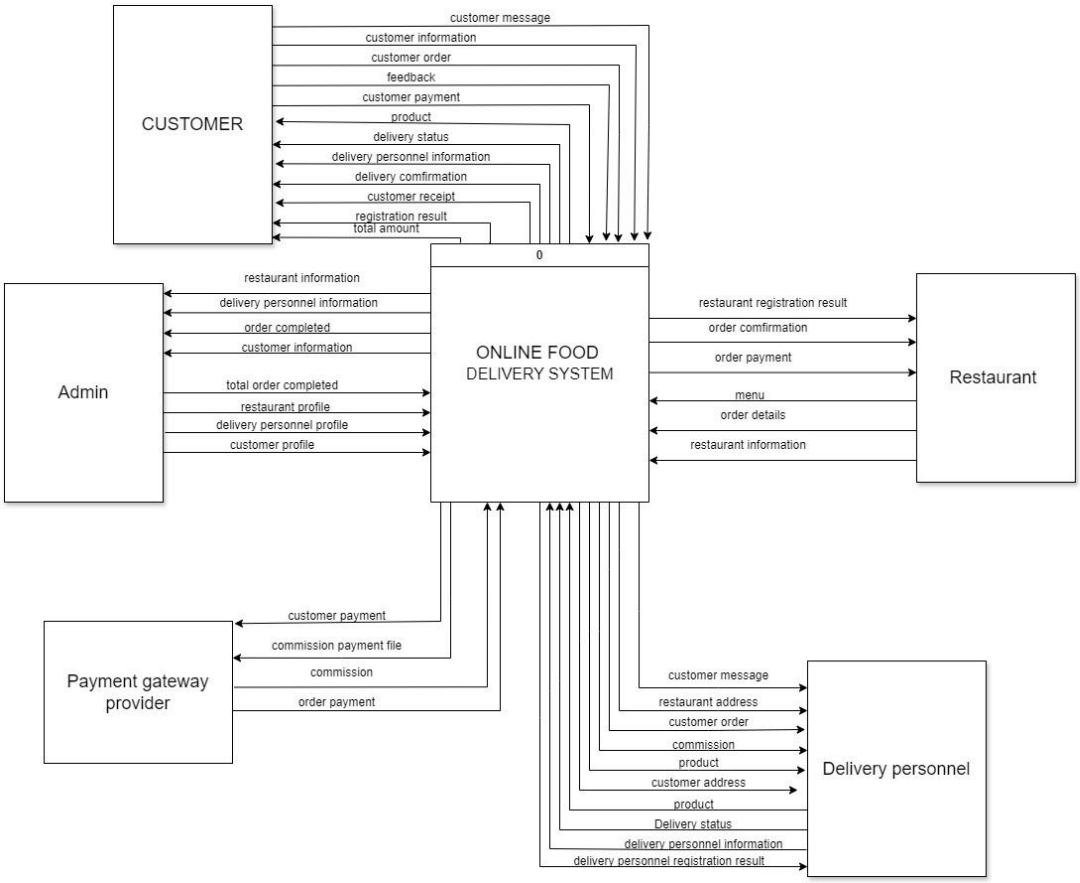


- Delivery Personnel live location page
- Delivery Personnel order completion page
- Delivery Personnel Commission page
- Delivery Personnel Commission details page

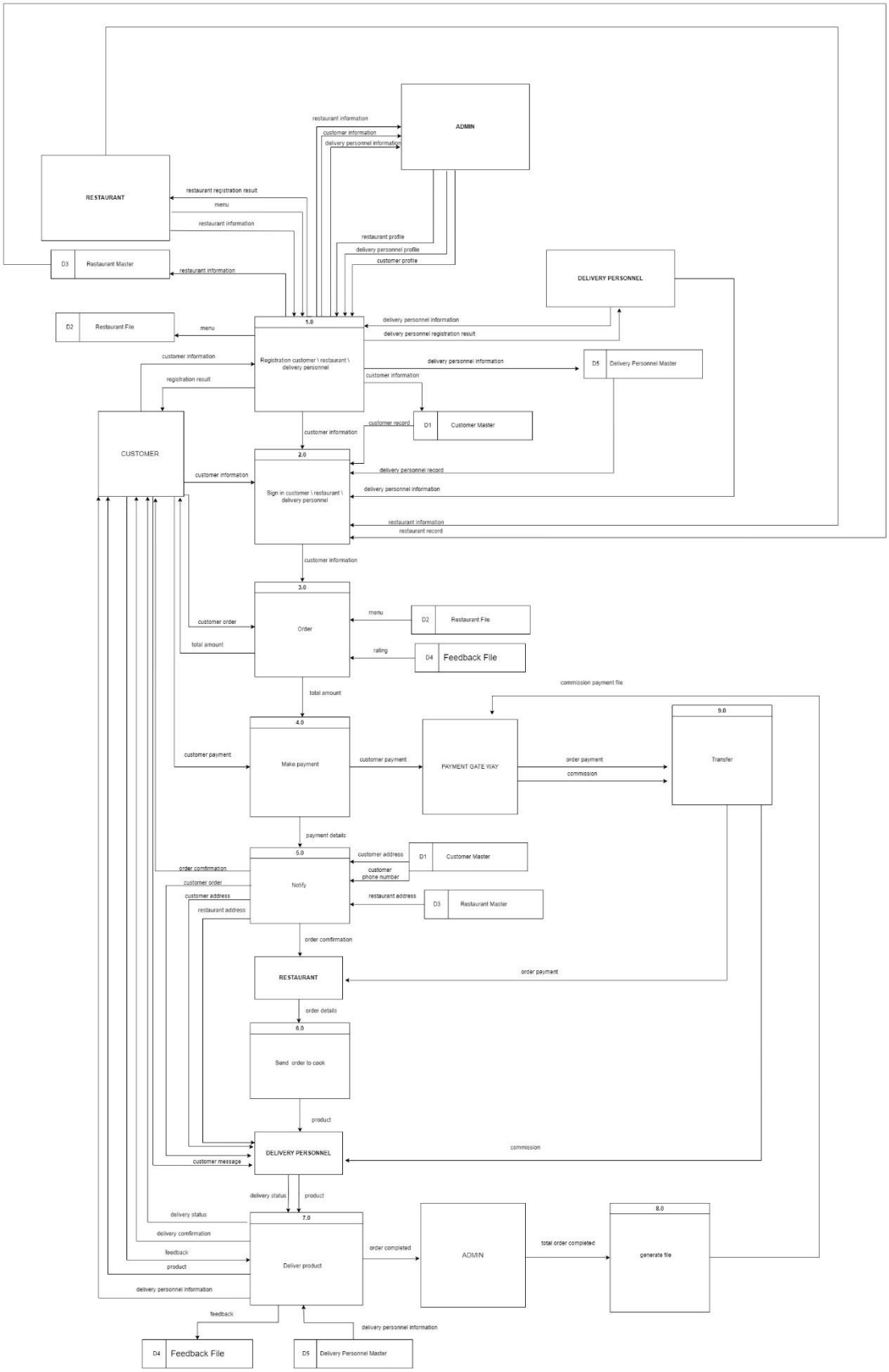


Updated Data Flow Diagram:

CONTEXT DIAGRAM

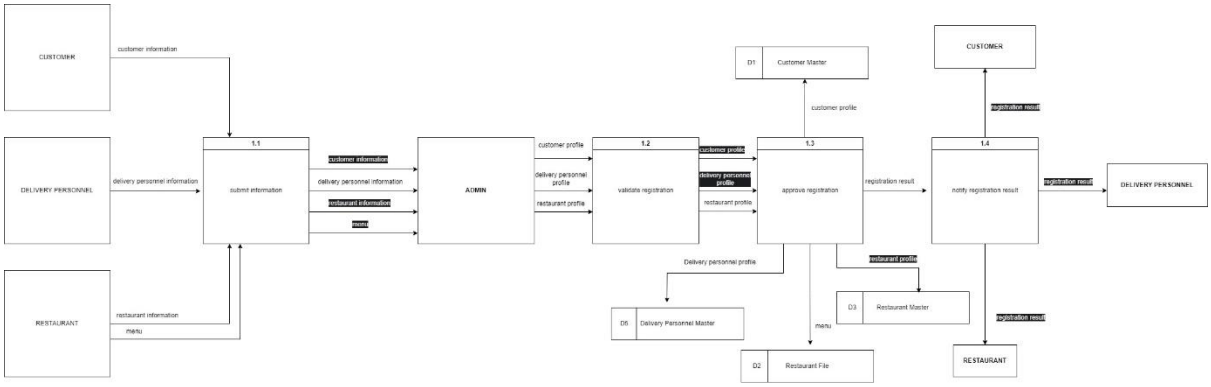


LVL 0

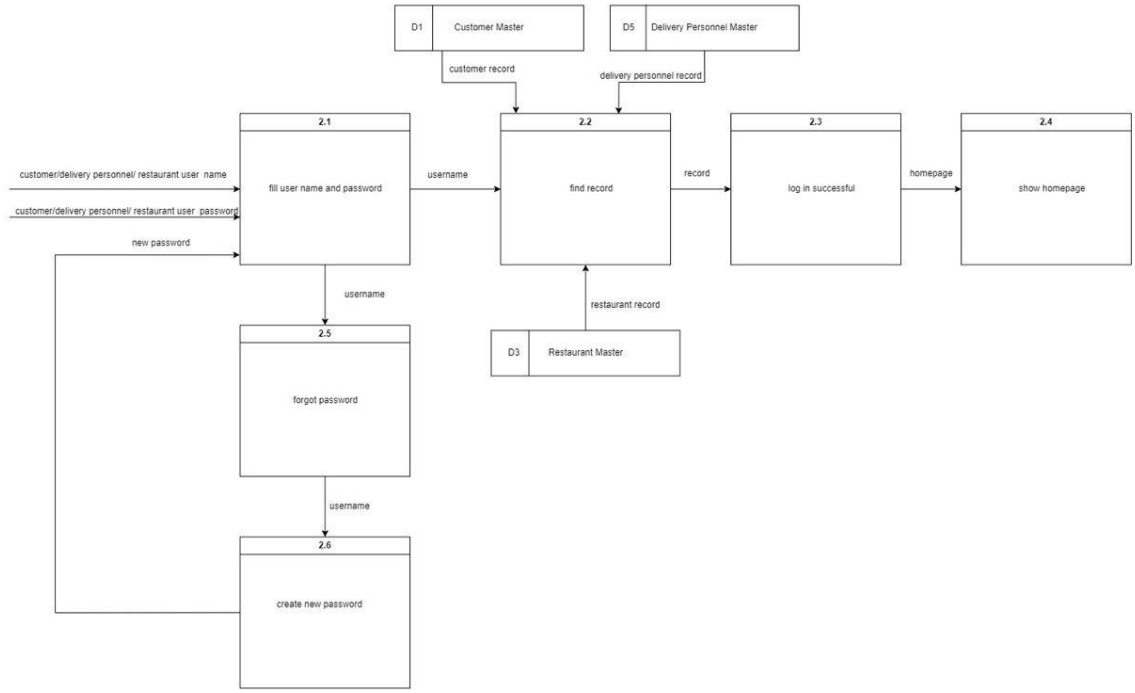

ADMIN
RESTAURANT
DELIVERY PERSONNEL
CUSTOMER
D1 Customer Master
D2 Restaurant File
D3 Restaurant Master
D4 Feedback File
D5 Delivery Personnel Master
1.0 Registration customer \ restaurant \ delivery personnel
2.0 Sign in customer \ restaurant \ delivery personnel
3.0 Order
4.0 Make payment
5.0 Notify
6.0 Send order to cook
7.0 Deliver product
8.0 generate file
9.0 Transfer
PAYMENT GATE WAY
restaurant information
customer information
delivery personnel information
restaurant registration result
menu
restaurant profile
delivery personnel profile
customer profile
delivery personnel registration result
registration result
customer record
delivery personnel record
restaurant record
customer order
total amount
rating
commission payment file
customer payment
order payment
commission
payment details
customer address
customer phone number
restaurant address
order confirmation
order details
product
customer message
delivery status
delivery confirmation
feedback
order completed
total order completed

LVL 1

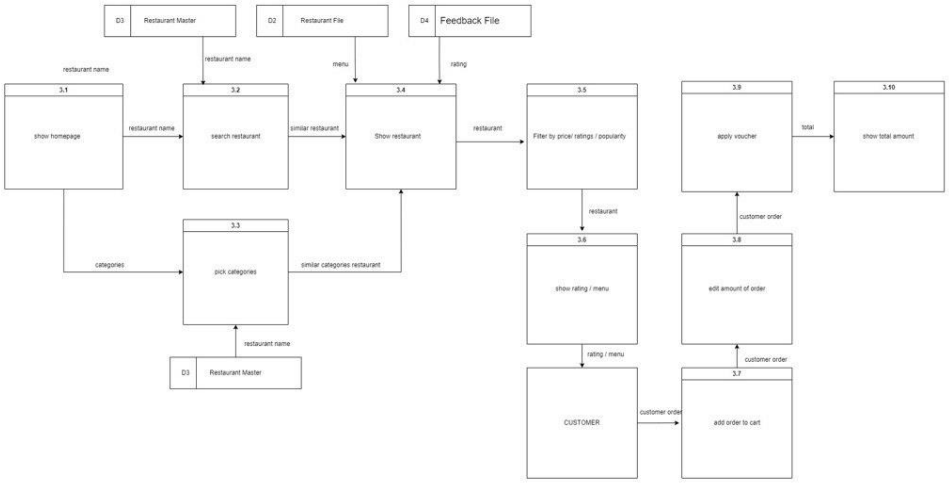
1.0


CUSTOMER
customer information
DELIVERY PERSONNEL
delivery personnel information
RESTAURANT
restaurant information
menu
1.1
submit information
customer information
delivery personnel information
restaurant information
menu
ADMIN
customer profile
delivery personnel profile
restaurant profile
1.2
validate registration
customer profile
delivery personnel profile
restaurant profile
1.3
approve registration
registration result
1.4
notify registration result
registration result
DELIVERY PERSONNEL
D1 Customer Master
customer profile
CUSTOMER
registration result
Delivery personnel profile
restaurant profile
D5 Delivery Personnel Master
D3 Restaurant Master
menu
D2 Restaurant File
registration result
RESTAURANT

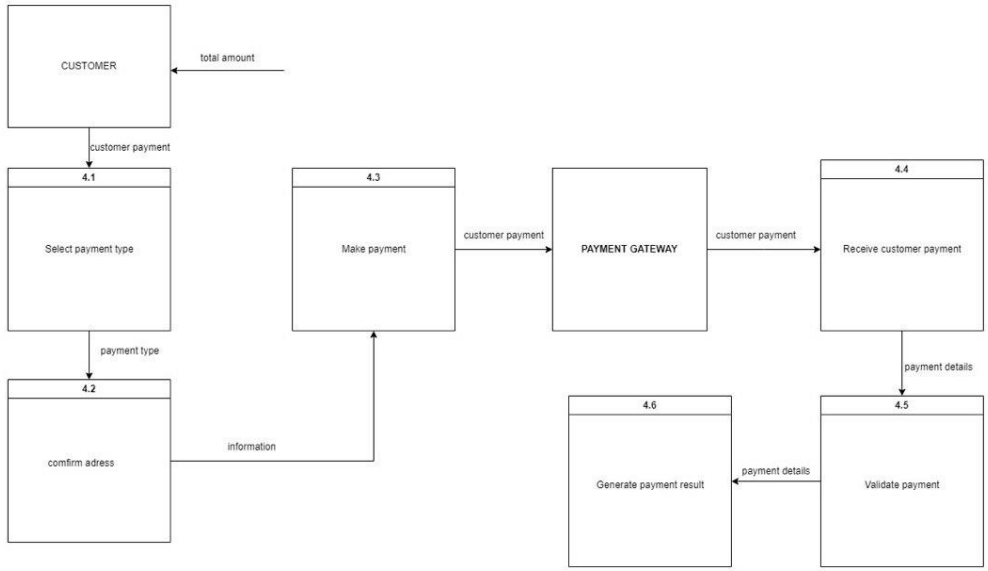
2.0


D1 Customer Master
D5 Delivery Personnel Master
customer record
delivery personnel record
2.1
fill user name and password
2.2
find record
2.3
log in successful
2.4
show homepage
customer/delivery personnel/ restaurant user name
customer/delivery personnel/ restaurant user password
new password
username
record
homepage
username
restaurant record
2.5
forgot password
D3 Restaurant Master
username
2.6
create new password

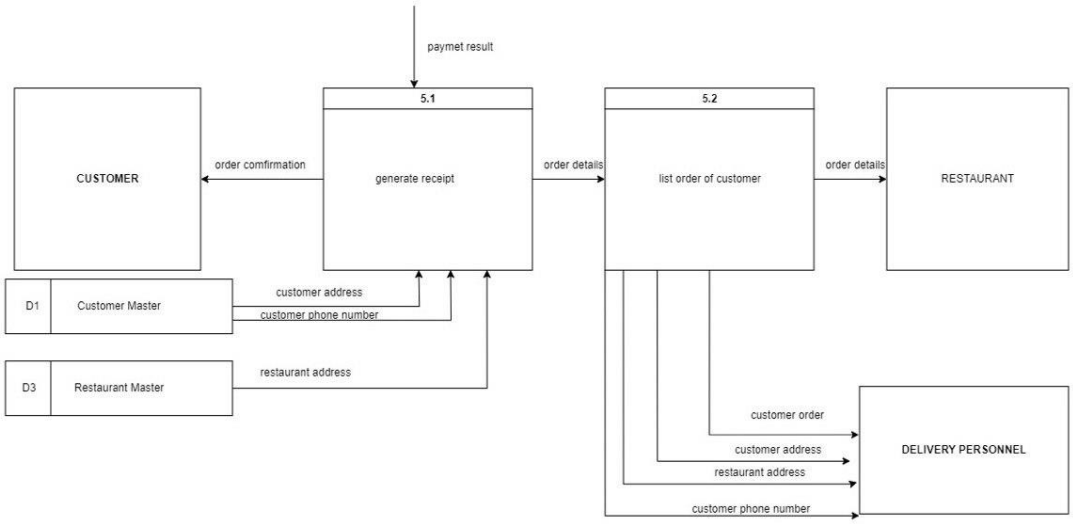
3.0


D3 Restaurant Master
D2 Restaurant File
D4 Feedback File
restaurant name
menu
rating
3.1
show homepage
3.2
search restaurant
3.4
Show restaurant
3.5
Filter by price/ ratings / popularity
3.9
apply voucher
3.10
show total amount
total
similar restaurant
restaurant
customer order
categories
3.3
pick categories
similar categories restaurant
3.6
show rating / menu
3.8
edit amount of order
rating / menu
customer order
D3 Restaurant Master
CUSTOMER
customer order
3.7
add order to cart

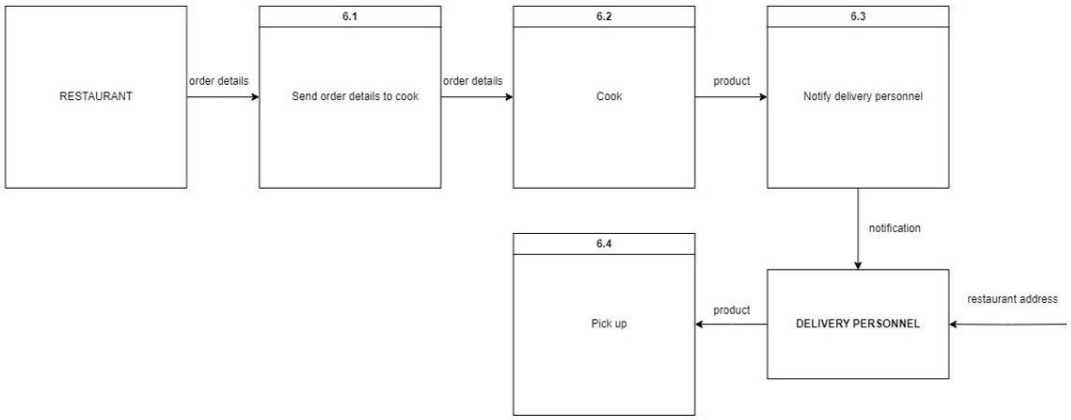
4.0


CUSTOMER
total amount
customer payment
4.1
Select payment type
4.3
Make payment
customer payment
PAYMENT GATEWAY
customer payment
4.4
Receive customer payment
payment type
payment details
4.2
comfirm adress
information
4.6
Generate payment result
payment details
4.5
Validate payment

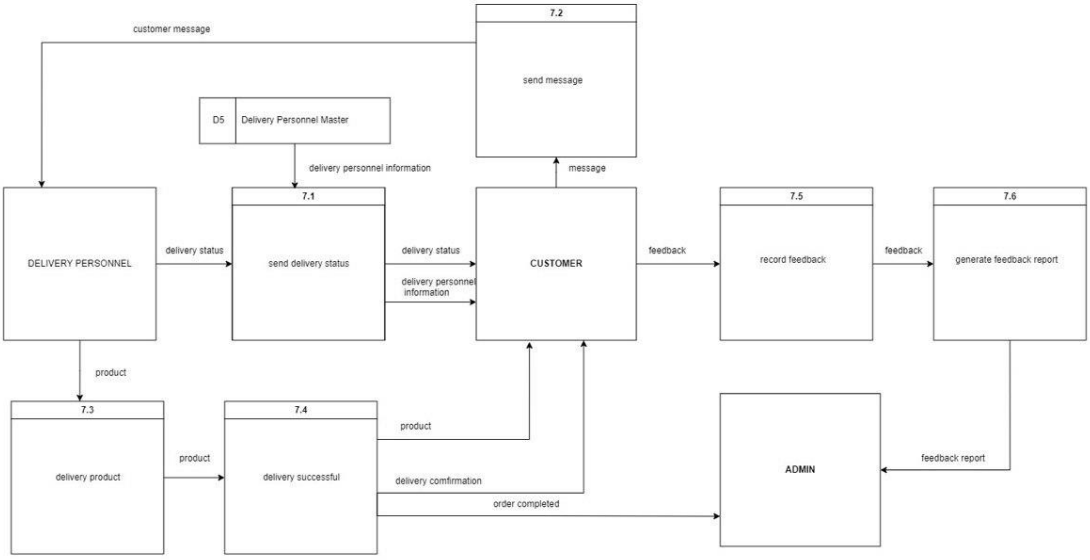
5.0


paymet result
5.1
5.2
CUSTOMER
order comfirmation
generate receipt
order details
list order of customer
order details
RESTAURANT
D1
Customer Master
customer address
customer phone number
D3
Restaurant Master
restaurant address
customer order
customer address
restaurant address
customer phone number
DELIVERY PERSONNEL

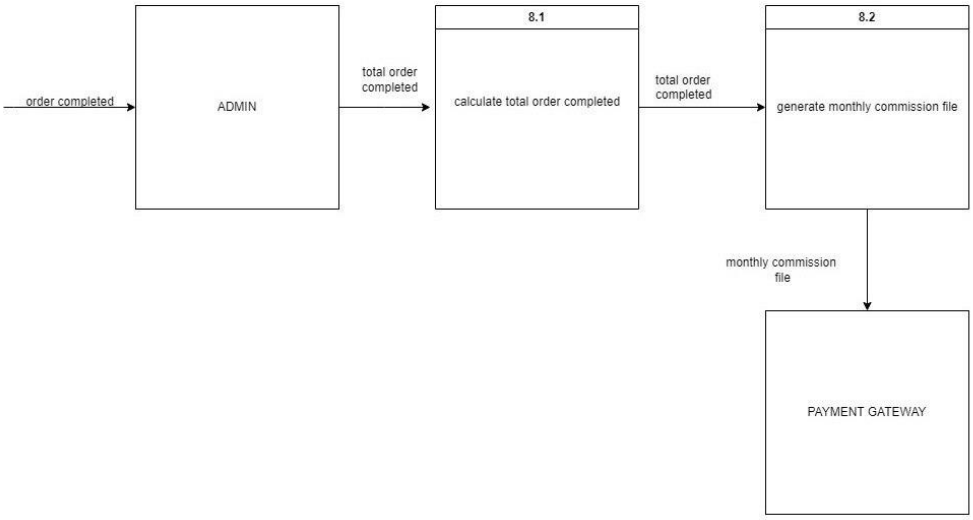
6.0


RESTAURANT
order details
6.1
Send order details to cook
order details
6.2
Cook
product
6.3
Notify delivery personnel
notification
6.4
Pick up
product
DELIVERY PERSONNEL
restaurant address

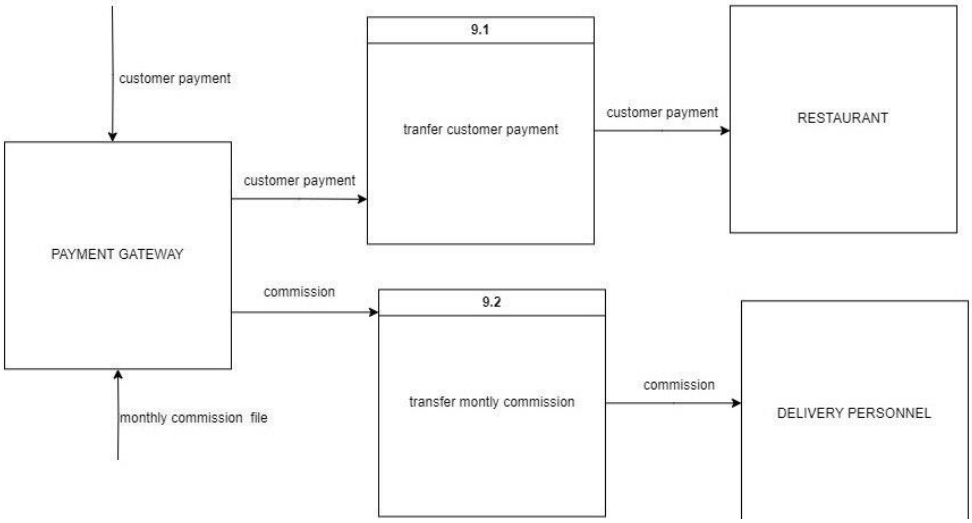
7.0


customer message
7.2
send message
D5
Delivery Personnel Master
delivery personnel information
message
DELIVERY PERSONNEL
delivery status
7.1
send delivery status
delivery status
delivery personnel information
CUSTOMER
feedback
7.5
record feedback
feedback
7.6
generate feedback report
product
7.3
delivery product
product
7.4
delivery successful
product
delivery comfirmation
order completed
ADMIN
feedback report

8.0


order completed
ADMIN
total order completed
8.1
calculate total order completed
total order completed
8.2
generate monthly commission file
monthly commission file
PAYMENT GATEWAY

9.0


customer payment
PAYMENT GATEWAY
customer payment
9.1
tranfer customer payment
customer payment
RESTAURANT
commission
9.2
transfer montly commission
commission
DELIVERY PERSONNEL
monthly commission file

DATA DICTIONARY

Data Store ID	Data Store Name	Data Store Description	Data Store Element				
			Name	Description	Base / Derived	Type of Data	Default value
D1	Customer Master	Keep all customer information	Customer Name	The name of customer	Base	Text	No
			Customer Address	The address of customer	Base	Text	No
			Customer phone number	The phone number of customer	Base	Text	No
			Customer order number	The order number of customer	Derived	Text	No
D2	Restaurant File	Stores details of each menu item	Menu Item Name	The name of the product	Base	Text	No
D3	Restaurant Master	Keep all restaurant information	Restaurant Name	The name of the restaurant	Base	Text	No
			Restaurant Address	The address of the restaurant	Base	Text	No

D4	Feedback File	Keep all feedback	Completed order description for restaurant	Users' feedback	Base	Text	No
D5	Delivery personnel Master	Keep all delivery personnel information	Delivery personnel name	The name of delivery personnel	Base	Text	No
			Delivery personnel phone number	The phone number of the delivery personnel	Base	Text	No

3.Conclusion

In this project, we developed the non-functional prototype interface of the system based on the previously drawn data flow diagram. Each process's user interface was meticulously designed, with comprehensive considerations for aspects such as error handling, system feedback, and pop-up notifications for users. We used tools to transform the design concepts into practical and operational interfaces, enabling people to clearly perceive how pages jump and how information flows, and simulating the operational logic of the system with great clarity. In the future, the interaction design can be improved by adding some dynamic elements and transition effects to make the user operation smoother and more intuitive.

For future, we could really step up the interface interaction design by throwing in more dynamic elements and transition effects. This would seriously boost the smoothness and intuitiveness of how users interact with it. It can be designed according to the responsive design to allow interface elements to adjust themselves, so that the user experience is more better.

Improve the user feedback mechanism to provide more detailed and personalized prompts and error messages, allowing users to know the operation results and solve problems by themselves. We need to provide more detailed and personalized prompts and error messages. For example, when giving error prompts, directly tell users where the error is and how to solve it, so that this will help users understand the operation results better and solve problems more easily, reducing their confusion and the chances of making mistakes.

We also have to beef up the data validation function. This ensures that the data users enter is accurate and complete. At the same time, we mustn't forget about system security. Using encryption and other techniques to protect user data is crucial. For example, we can validate key data in real-time to stop illegal data from being entered or stored. This will keep the system running smoothly and safeguard users' information.

4. Proof of discussion

